Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **C++** and **Python** systems, concurrent data-processing workflows, automation, and AI-enabled applications using **algorithms**, Linux, cloud infrastructure, performance optimization, automated testing, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Operating Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Data Retrieval, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Architected a multi-tenant platform using Azure SQL and Blob Storage with row-level security and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built FastAPI integrations across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, error handling, and automated CI/CD releases sustaining 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Designed concurrent processing and secure multi-format ingestion through REST APIs and AWS RDS, executing 1K+ files per batch with fault-tolerant recovery, usage tracking, and stable performance under load.
- Built Python services and responsive React interfaces for uploads, history, configuration, and live feedback, reducing user turnaround 60% while supporting reliable production workflows used by 150+ instructors.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Designed object-oriented API integrations and AWS Glue transformations for JSON, CSV, and Parquet data, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20%.
- Automated nightly builds, installations, unit tests, integration tests, and reports through Jenkins, validating reliable cross-platform synchronization while collaborating directly with Microsoft and GitHub stakeholders throughout Agile delivery.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight MERN microservices for authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across production workflows.
- Built responsive React workflows and automated Docker deployments through Jenkins CI/CD, achieving 95%+ build validation while integrating operational monitoring and analytics across controlled staging and production environments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built cross-platform mobile and web booking workflows with Flask APIs, Firebase synchronization, payments, and webhooks, processing hourly Databricks micro-batches across 50K+ records with consistently reliable sub-200-millisecond responses.
- Designed end-to-end workflows, mapped 12+ integration points, and presented technical solutions to client stakeholders, reducing design errors 25% while supporting effective collaborative cross-functional implementation planning and alignment.

Skills

Programming & Fundamentals: C++, Python, Java, Object-Oriented Programming, Data Structures, Algorithms, Operating Systems, Concurrent Processing, Performance Optimization
AI & Automation: PyTorch, TensorFlow, Transformers, LLM Systems, Feature Engineering, Hyperparameter Tuning, Automation, Scripting
Systems & Cloud: Linux, Docker, AWS, Azure, GCP, Kubernetes, Distributed Systems, REST APIs, Monitoring, Fault Tolerance
Engineering: Git, GitHub, Jenkins, CI/CD, Unit Testing, Integration Testing, Debugging, Code Reviews, Documentation, Agile/Scrum

Projects

CPU Scheduling Simulator Live | *C++, Operating Systems, Algorithms, React.js*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multi-level queue scheduling with interactive controls and Gantt visualization, rigorously testing correctness across 1K+ preemptive and non-preemptive process combinations.

CaptionCraft - Image Caption Generator Live | *PyTorch, Transformers, Flask, Deep Learning*

- Benchmarked BLIP, GIT, and ViT-GPT-2 transformer architectures across 10K images and 100K captions, improving measured caption accuracy from 81% to 96% while building reliable scalable inference APIs.

Linguistix | *Python, AWS, Docker, CodePipeline, ECS Fargate*

- Developed and deployed a containerized RESTful data-processing API on AWS using CodePipeline and ECS Fargate with automated releases, validation checks, 87% accuracy, and dependable production-grade operational reliability.